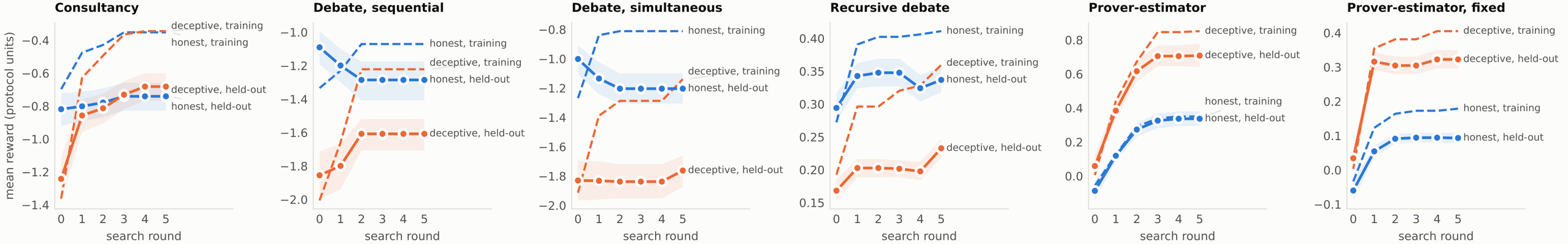
Optimizing against the judge pays off for lying, not for honesty

Best strategy so far in each search, 3 judges pooled. Dashed: reward on the 36 training items it was selected on (the proxy the search climbs). Solid, with 95% item-bootstrap band: reward on 80 held-out items.